

1 Local configuration file ctaipplay.ini

The layout is configured for specific terminals in the file ctaipplay.ini in the terminal directory c:\ctaip.

This file is used for the configuration of grids in AIP.

The server directory \hydra\ctnet\win\aip includes complete INI files belonging to the HYDRA standard. Any deviations are developed in specific, customized directories e.g. \hydra\1\custom\aip\tgrp_901.



Create the corresponding, empty file (e.g. ctaipplay.ini) in this directory. Modified sections are copied to this file. Make the configurations in this file.

After restarting the terminal, files from the main directory \hydra\ctnet\win\aip are merged with files from the customized directory \hydra\1\custom\aip\tgrp_901. Then the merged file is transferred to the local terminal directory C:\aip.



Changes to the configuration file ctaipplay.ini will not take effect until the terminal software has been restarted.

Entry	Comment
Section [OP info]	
Deaktiviert=AG_Bmk,AG_Fort	- Indicated info pages are not shown - AG_Info (OP info) cannot be disabled - entries affected by sorting are not disabled.
Sortierung=AG_TechInfo,*	- Order of info pages in the icon list - if the list ends with " , * " the non-listed standard pages are added at the end. - Standard pages: AG_Info,AG_ZuInfo,AG_Bmk,AG_TechInfo,AG_Fort,AG_FertP ap
Sortierung=AG_Info,AG_AD_Satz,Text File1,TextFile2,Bild1,* TextFile1=C:\LOGOPAK\druck.txt,druck .txt TextFile2=C:\LOGOPAK\druckerg.txt,dr uckerg.txt Bild1=C:\aip\spool\screen1.bmp	Configuration of any texts and images that can be displayed in the OP info dialog.
Section [main]	
Nachkommastellen=0	Decimal places for quantities in the order/machine overview
Repaint_time=60	Cycle for updating the view (for machine list and machine info)
Buttonkonfigaktiv=0	reserved

Entry	Comment
layout=100	reserved
AuftragZusatzInfoFontSize=12	Font size of texts in the "OP info" dialog
AuftragZusatzInfoFontName=Courier New	Font type of texts in the "OP info" dialog
PopupSize->EmptyQueue=300	Empty popup window size for quick queue
PopupSize->ReloadPze=200	Reload popup window size for PZE configuration
SymbolAdditionalInfo=MBEZK	Display of any field from the machine list in the icon view between machine number and operation number The machine number is replaced by the configured field if lines and aggregates are configured.
SymbolSubstDesignation=MBEZK	The specified field replaces the machine number in the icon view.
Sections for list layouts	
[Personenliste]	List displayed when staff is logged on
[Bedienposition]	Predefined list of "operator positions"
[Maschinenstatusliste]	Predefined list of "machine statuses"
[Ausschussgruende]	Predefined list of "scrap reasons"
[Abweichungsgruende]	Predefined list of "deviation reasons"
[Auftragsliste]	Lower list in the main view
[Vorgabeliste]	Order sequencing list
[Schichtinfo]	List of shift info
[Maschinenliste]	Upper list in the main view
[Eingangslsliste] input batch list	List of input batches, e.g. when "logging on the OP" in batch mode
[Ausgangslsliste]	List of preceding batches when "changing output batches"
Syntax of table formatting	
GRID_FONT=Arial	Font type
GRID_FONTSIZE=9	Font size
GRID_COLOR=clBlack	Font Color
GRID_BACKGROUND=clWhite	Background color
GRID_FIXCOLS=0	Number of fixed columns
GRID_ORDER=MSZEIB GRID_ORDER=MSZEIB=-	Sorting Sorting in descending order Please note: If several criteria are indicated (separated by), only the first criterion can be sorted in descending order. All other criteria are sorted in ascending order. The following entry must be set in the configuration for the section so that the sorting is used in the display: ORDER=#USE#INI#ITEM# <u>Example for the section Sequencing List (Auto)</u> [WF@ANR] CMD=DLG=LIST;11 MOD=V MNR=<MNR> SECTION=Sequencing List (Auto) ORDER=#USE#INI#ITEM#
GRID_FILTER=MGRP=0	Table filtering Please note: NOT supported for script dialogs.

Entry	Comment
GRID_LIST_TYP=MNR GRID_LIST_TYP=ANR	The list type of the section is specified with this entry, if fields are displayed that need to be loaded additionally. This entry also enables the search when starting. The entry has to be entered above the IDs to be reloaded!!! All IDs that can be reloaded can be found in the file headers.txt in the "spool" directory of the terminal. It consists of four lines: 1. Fields that are always included in the machine list 2. Fields that can be reloaded for the machine list 3. Fields that are always included in the order list 4. Fields that can be reloaded for the order list
EXAMINE_CONTENTS_CHANGE=MGRP EXAMINE_COLOR_C1=clWhite EXAMINE_COLOR_C2=clSilver	The font color switches from clWhite to clSilver every time the MGRP value changes. Up to 8 colors can be defined.
EXAMINE_SCANEXPR1=MGRP=71 72 73 EXAMINE_SCANEXPR2=MGRP=96 97 101 EXAMINE_SCANCOLOR1=clGreen EXAMINE_SCANCOLOR2=clRed	The machine groups 71/72/73 are presented in green font color; the groups 96/97/101 are displayed in red font color. Up to 8 colors each can be defined.
EXAMINE_SCANBKEXPR1=BATTRIB=1 EXAMINE_SCANBKEXPR2=BATTRIB=2 EXAMINE_SCANBKCOLOR1=clBlue EXAMINE_SCANBKCOLOR2=clLime	All lines with BATTRIB=1 are shown in blue background color; rows with BATTRIB=2 are displayed in lime. Up to 8 colors each can be defined.
EXAMINE_COLOR=TEXTCOLOR EXAMINE_BKCOLOR=HGRCOLOR	Specification of a column that includes the color value for the row (e.g.: 0-Black; 255-Red, 16777215-White)
EXAMINE_CELLBKLEVEL2=EGR:AUSP,EGR:AUSP,<1*clLime <=5*clYellow >15*clRed	Setting of the background color depends on whether the field value reaches different threshold values.
EXAMINE_ROWBKLEVEL1=SGR:REST,SGR:REST,<=0*clRed <=5*clLime >15*clYellow	Setting of the background color depends on whether the field value reaches different threshold values. The entire row is colored because of the threshold value.
Syntax of column definitions	
MNR=C8,80,R MGRP=N6,60,R AGR:AUS=N10.2,125,R MSDATB=dd.mm.yy,70,L MSZEIB=hh:mm,70,L SKDATB=dd.mm.yyyy,90,L SKZEIB=hh:mm:ss,80,L MSDAUER=ddd.iii,60,R	Alpha-numeric, 8 characters, 80 pixels, right-aligned Numeric, 6 characters, 60 pixels, right-aligned Decimal, 10 digits, 2 decimal places, .. Displayed in the form "23.03.98" (left-aligned) Displayed in the form "08:24" Displayed in the form "23.03.1998" Displayed in the form "08:24:39" Displayed in industrial time unit " 22,982"
AGR:BMK11=hhh:mm:ss,80,R,TESTHEADER	TESTHEADER: new column caption
ALIAS KOPIE=MNR=N8,120,R,TITEL	ALIAS new name is being introduced KOPIE= new identification MNR= ID in data file N8,120,R, Formatting TITEL column caption in table
ALIAS AKA=MNR[1..3]=N8,120,R,ARRAY[1..3]	The first three characters from MNR are displayed.

Entry	Comment
ALIAS ATTR=MNR(2)=N8,120,R,PARAMETER(2)	The second part separated by „ ; “ is displayed. Example: „ 12;20;130 “ → „20“
ALIAS U=(R*6.2831)=N10.3,60,L,Scope	Conversion of a value Syntax: see below
ALIAS SOLLZ={_INT((_DATETIME(SKDATE , SKZEIE)-_DATETIME(SKDATB , SKZEIB))*86400)}=hh:mm:ss,60,R,SOLLZ ;target time of shift	Complex calculations relating to several fields Syntax: see below
GRID_BROWSEROW=0	only the active row is colored yellow
GRID_CELLPAINT=ON	Requirements for coloring rows column by column
GRID_REFRESH=5000	Cycle for updating the display [ms] → lists are not reloaded from the server! Recommended if a constantly changing value is calculated using an ALIAS function.
GRID_POSITION=ON	Display of the grid position  Limited/no support when scrolling using scroll bars and page scrolling
GRID_CELLPAINT=ON EXAMINE_CELLBKCOLOR=WTK:STA,WTK:STA,0-clGreen 1-clBlue 2-clYellow 3-clRed can also be used with index: EXAMINE_CELLBKCOLOR1..8 As of 2.0.2.64: EXAMINE_CELLBKCOLOR1..20	One single column is colored in every cell subject to a value. 1st value: ID of the column to be colored. 2nd value: ID of the reference column 3rd value: Configuration (color for possible values) Notes: - The reference column MUST be shown in the list, if required with length 0 - The values are converted into capital letters when being compared.
As of ctaip V# 2.0.2.17 EXAMINE_CELLBKCOLOR=DMY,COLOR	Take over the color directly from the "color" column. The column <DMY> is shown in the color defined in the column <COLOR>

Entry	Comment
; Definition virt. column(1) EXAMINE_CELLVALUE1=CV1,REF1,S =DAT P=ZEI A=REST N=INFO ; Definition virt. column(2) EXAMINE_CELLVALUE2=CV2,REF2,1 0=MGRP 20=MNR 30=COLOR 40=MS TTXT ... ; Layout/Position virt. column(1) CV1=CELLVALUE,150,Z,Data ... ; Layout/Position virt. column(2) CV2=CELLVALUE,150,Z,M/C/T	Filling of a virtual "Case" column with values from different columns subject to the value of a reference column. 1rd value: Identification of the virtual "case" column 2rd value: Identification of the reference column 3rd value: Configuration Reference value + ,=' + display column Please note: the virtual "case" column needs to be configured as follows <Identifier>=<Key word>,<Width>,<Alignment>,<Caption> e.g. CV1=CELLVALUE,150,Z,YST
GRID_RANDOMSORT=ON	This options randomly sorts the list. Please note: If this option is active, any configured sorting will be ignored.
GRID_CLIPBOARD=<BUTTON>@<SELECT>@<DATA>	This option copies data from a table/grid into the clipboard.
Special entries	
[Auftragsliste] ALIAS Restlaufzeit=RLZ= C4,44,R,RLZ ALIAS SollErreicht=ZBV1= C1,12,R,S	Activation of the calculation & display of the remaining run time Activation of the calculation & display for target quantity reached (column 'S': '*' if reached)
[Maschinenliste] ALIAS StkProMin=IZYSM= N8,48,R,Stk/min	Activation of calculation & display of the produced pieces per minute
[layout pze] KundenBitmap=kunde.bmp DienstGangTaste=1,3 StdSchrift=Arial StdDateSize=30 StdStatusSize=26 StdSpdBttSize=16 InfoSchrift=Courier New InfoSchriftSize=20 SmallStatusFontSize=16 DateTimeLayout=dd.mm.yyy hh:mm:ss	Configuration of the PZE terminal "Kundenbitmap=<File name>" file with customer logo When restarting the terminal, this file is copied from the server directory ".\ctnet\win\aipl\etc\" into the application directory ".\etc\". „DienstGangTaste=1,3“ → Default [empty] By entering the function key numbers (1...4), a check specifying if the person is allowed to go on a business trip is performed during the posting. Configuration of the used font types/font sizes as well as the layout of the date and time display.

1.1 Formulas used in grid layout

Simple conversion of a value

Syntax: **ALIAS** <Alias>=(<formula>)=formatting

<formula>: [1/]<ID>[<Operator><Value2>]

<ID>: ID from list (The current value from the list is entered here
in the formula)

<Operator>: + | - | * | / | ^

<Value2>: 2nd Operand

Extensive formulas:

Formulas that can also relate to several table fields can be recognized by braces.

Syntax: **ALIAS** <Alias>={<Formula>}

<Formula>: (<Operand1>[<Operator><Operand2>])

<Operand>: <Value> / <Function> / <Formula> / |KENN|

<Operator>: |+|-|*|/|^|

<Value>: Constant ('0'..'9','e','E','.', '-')

|Kenn|: reads out a value from the table

<Function>: _<Fname>(<Operand>[,<Operand>[,...]])

- _ **DATETIME** (<Date>,<Time>)
 - <Date>: mm/dd/yyyy
 - <Time>: ssss (Seconds of the day (0..86400))
 - ==> Real value as TDateTime
- _ **INT** (<Operand>)
 - returns a value without decimal places
- _ **REAL** (<Operand>)
 - returns a value with decimal places
- _ **ROUND** (<Operand>)
 - returns rounded value without decimal places
- _ **MOD** (<Operand1>,<Operand2>)
 - ==> Operand1 mod Operand2
- _ **ABS** (<Operand>)
 - absolute amount of a figure
- _ **EXP** (<Operand>)
 - e raised to the power of X (e: basis of the natural logarithm)
- _ **LN** (<Operand>)
 - natural logarithm (Ln(e) = 1)
- _ **FRAC** (<Operand>)
 - proportion of decimal places
- _ **LOG** (<Operand1>,<Operand2>)
 - LOG(N,X): logarithm to base N of X
- _ **MAX** (<Operand1>,<Operand2>)
 - the greater value of two values
- _ **MIN** (<Operand1>,<Operand2>)
 - the lesser value of two values
- _ **SQRT** (<Operand>)
 - ==> Square root of Operand1
- _ **PI** ()
 - ==> 3.14151926535...

The option "GRID_BROWSEROW=0" should also be set in order for the coloring to be recognized even if the row is selected.

Examples:

```
EXAMINE_CELLBKLEVEL1=MNR,MST,<=1*c1Lime|<=2*c1Yellow|>2*c1Red
EXAMINE_CELLBKLEVEL2=FS,FS,<90*c1Lime|>=90*c1Yellow|>=100*c1Red
EXAMINE_CELLBKLEVEL3=EGR:GUT,EGR:GUT,<(SGR:GUT)*c1Lime|>=(SGR:GUT)*c1Yellow
```

1.2 Translations in grid layout

Column contents can be configured to be translated and displayed by entering e.g. the configuration <XYZ=T10,100,L> instead of <XYZ=C10,100,L> in the configured grid columns. A <#> character must be prefixed for these "resource strings" to provide for better classification. This modification can be used in every INI file (hytnrcfg.ini,..) where grid layouts are configured.

Please note: The data do not include any translated values. In order for them to be displayed in e.g. dynamic dialog fields, an explicit translation must be performed using the VB script function <vbsTranslateDataValues("<columns>" , "<data row>") >.

Column contents can be configured to be translated and displayed by entering e.g. the configuration <PSPERRE=U1,100,L> instead of <PSPERRE=C1,100,L> in the configured grid columns. The entry for the "resource string" that depends on the field has the following structure:

	„#<Acronym>#<Value>“	
e.g.	„#PSPERRE#J“	"production lock enabled"
	„#PSPERRE#N“	„ “ (blank character)

This modification can be used in every INI file (hytnrcfg.ini,..) where grid layouts are configured.

Please note: The data do not include any translated values. In order for them to be displayed in e.g. dynamic dialog fields, an explicit translation must be performed using the VB script function vbsTranslateDataFields("<columns>" , "<data row>") >.

1.3 Configuration of basic screens

The dialogs/screens are configured using dynamic dialogs. For this reason, the following dialogs are always required:

MMINFO → Section referring to machines in the single machine view

MAINFO → Section referring to orders in the single machine view

60610		60611		60612		60614		NA43		GAP1	
Arbeitsplatz/Maschine	60612	Status	PRODUKTION								
Bezeichnung	ARB-320S	Beginn	14.08.10.2009								
Gruppe	SGM2	Dauer [Std:Min]	20:01								
Teiligkeit	1	Gutmenge	0.000000 ST								
Sollzyklus [Sek/Takt]	21	Ausschuss	0.000000 ST								
Istzyklus [Sek/Takt]	0.00	Takte	0.000								
Information anzeigen Produktionsstatus sperren Status ändern Arbeitsgang anmelden											
Arbeitsgang	S61003310100	Sollmenge	60000.000 ST								
Artikel	SAL-FMO011-K-	Gutmenge	228.000000 ST								
Artikelbezeichnung	Spiegelgehäuse	Ausschuss	4.000000 ST								
Bemerkung 1		Soll seit Anm.	--- ST								
Bemerkung 2		Gut seit Anm.	0.000000 ST								
Plandauer	350:00	Abweichung [%]	---								
		Fertigstellung	0%								
Information anzeigen Teiltrückmeldung Arbeitsgang unterbrechen Arbeitsgang abmelden Hilfsmittel											

CTAIP V1.2.4.64 29.09.09 Server:10.10.60.70:3 TNR:126 Status: ON 09.10.09 10:02:57

MINFO → Description of the machine information

Maschine 60612 Arbung Selogica 320S	
BDE-Kommentare	Angemeldete Personen
Arbeitsplatz/Maschine	60612
Bezeichnung	ARB-320S
Gruppe	SGM2
Teiligkeit	1
Sollzyklus [Sek/Takt]	21
Istzyklus [Sek/Takt]	
Status	PRODUKTION
Beginn	14.08.10.2009
Dauer [Std:Min]	20:03
Gutmenge	0.000000 ST
Ausschuss	0.000000 ST
Takte	0.000
Maschineninformation schließen	

AINFO → Description of the order information

The heights of the individual components of the basic screens and, as a result, the positions of the button bar are configured in the *ctaipplay.ini* file using the below-mentioned parameters:

Section [MainView1]	Configuration of the basic screen
MachineGridHeight=415 OrderGridHeight=500 ButtonBarHeight=50	Height configuration of components for the basic screen (machines, order grid, button bar) The configured heights are scaled to the current height. Consequently, the total sum of entered heights does not play a role.
Section [MainView2]	Configuration of the single machine view
MachineGridHeight=50 MachineInfoHeight=415 OrderInfoHeight=355 ButtonBarHeight=50	Single-row grid to select the machine Information on the machine Information on the order Height of both button bars The configured heights are scaled to the current height. Consequently, the total sum of entered heights does not play a role.

1.3.1 Available fields for the dialog configuration of basic screens

A script function that fills the fields according to the customer's requirements is currently not available (14 September 2009).

In general, the fields of the machine list and the order list are available. "MNR." or "ANR." must be prefixed for identification purposes.

Known quantity fields are formatted to match the configured number of decimal places.

Some fields are calculated. The following fields are additionally available:

Acronym	Description
ANR.SOLL_SEIT	Target quantity since login The value is determined locally at the terminal. This is only useful for MDE machines. However, the order must be logged on locally after restarting the terminal.
ANR.ABWEICH	Deviation [%] Comparison of "target quantity since logon" and "actual quantity since logon"
MNR.SZY	Target cycle As of AIP 2.0.2.85: Field is transferred including "internal decimal places". The number of characters displayed is determined by the field of the dialog configuration.
MDE.IZY	Actual cycle The machine's current actual cycle - only if MDE processing is active for the machine at this terminal.
MNR.MSZEIB	Start time of the current status
MNR.MSDATB	Start time of the current status
MNR.MSDAUER	Duration of the current status
ANR.BEARBZ	Planned duration
MNR.MSTTXT	Status text
MNR.TLG	Partitioning Calculated based on the orders running at the machine.
ANR.FERTIG	Progress bar
TNRSPERRE	Translated text for the production lock (corresponds to the configuration "TNRSPERRE=U1,150,L,Hinweis" in ctaiply.ini) (the value J/N from the list can be found in MNR.TNRSPERRE)

1.4 Integrate dynamic dialogs in information dialog

Dynamic dialogs can be integrated in information dialogs. The toolbar of the information dialog is then shown below the toolbar of the dialog:

Maschine 60610 Engel

BDE-Kommentare	Angemeldete Personen	Statusprotokoll	Beschreibung
Auftrag Arbeitsplatz 60610			
Datum/Zeit Maschine Kommentar	[07.10.2009 09:34:26] [60612] [444444444444]	Person Auftrag [S61003310100] [00050014 Albert,Claudia]	
Datum/Zeit Maschine Kommentar	[07.10.2009 09:33:47] [60612] [werqwerqwerwer]	Person Auftrag [S61003310100] [00040563 Griebner,Maria]	
Datum/Zeit Maschine Kommentar	[07.10.2009 09:33:21] [60612] [Kommentar 2]	Person Auftrag [S61003310100] [00040256 Pernikova,Lisa]	
Datum/Zeit Maschine Kommentar	[07.10.2009 09:32:54] [60612] [Kommentar (in ctwin erstellt)]	Person Auftrag [S61003310100] [00000667 Schulz,Christian]	
Datum/Zeit Maschine Kommentar	[21.01.2009 17:12:47] [LACK02] [FARBE LEICHT ABWEICHEND]	Person Auftrag [S30002000300] [00000667 Schulz,Christian]	
Datum/Zeit	[21.01.2009 14:26:07]	Auftrag [S30002000200]	
Seite 1 • Seite 2			
Abbrechen		Kommentar verfassen	
Maschineninformation schliessen			

Definition of a dialog for the machine info (ctaipplay.ini):

[M-Info]

;;Dialog<n>=<DLG>,<Tab

Caption>

Dialog1=WF_BDE_KOM_CHK,BDE

comments

Sortierung=Dialog1,*

Up to 10 dialogs can be configured (Dialog1...Dialog10). The entry "Sortierung=..." causes the tab with the new dialog to be put at the beginning.

Configuration of the toolbar of the info dialog for the dialog (ctaipbut.ini):

[M_INFO.DIALOG1-Page1]

1=MI_CLOSE,L,close machine information

If the "cancel" key is defined in the dialog as described in the above example, it will have no effect here.

It must be noted that the tabs are displayed over several rows if too many tabs have been inserted. Consequently, the tabs are moved downwards. Relevant areas might reach beyond the display area. To avoid this, the configurable caption texts of the tabs should be as short as possible.

1.4.1.1.1 Restrictions

A dynamic dialog is generated, once it is called. Output data (machine, order) are already available.

The dialogs integrated in the information dialogs are already set up when starting the application. Consequently, placeholders included in text fields (e.g. machine <MNR> <MBEZK>) cannot be converted. The function cannot just be started later, as the placeholders have already been overwritten during initialization.

1.5 TextViewer in dynamic dialogs

BDE-Kommentare

Auftrag S30008880100 Arbeitsplatz 60614

Datum/Zeit	Maschine	Person	Auftrag
[07.10.2009 09:34:26]	[60612]	[00050014 Albert,Claudia]	[S61003310100]
Kommentar [444444444444]			
[07.10.2009 09:33:47]	[60612]	[00040563 Griebner,Maria]	[S61003310100]
Kommentar [werqwerqwerwer]			
[07.10.2009 09:33:21]	[60612]	[00040256 Pernikova,Lisa]	[S61003310100]
Kommentar [Kommentar 2]			
[07.10.2009 09:32:54]	[60612]	[00000667 Schulz,Christian]	[S61003310100]
Kommentar [Kommentar (in ctwin erstellt)]			
[21.01.2009 17:12:47]	[LACK02]	[00000667 Schulz,Christian]	[S30002000300]
Kommentar [FARBE LEICHT ABWEICHEND]			
[21.01.2009 14:26:07]			[S30002000200]

Abbrechen Kommentar verfassen

Dialog configuration:

"General" tab:

Positioning by field X,Y

Size by unit X,Y

Felder ändern

Dialog **WF_BDE_KOM_CHK** Typ **AIPDEF** Dlg-User **0** Verweis **4341**

Allgemein | **Format** | **Funktionen** | **Optionen**

Aktiv an:
☒ immer
☐ nur Konsole
☐ nur Terminal

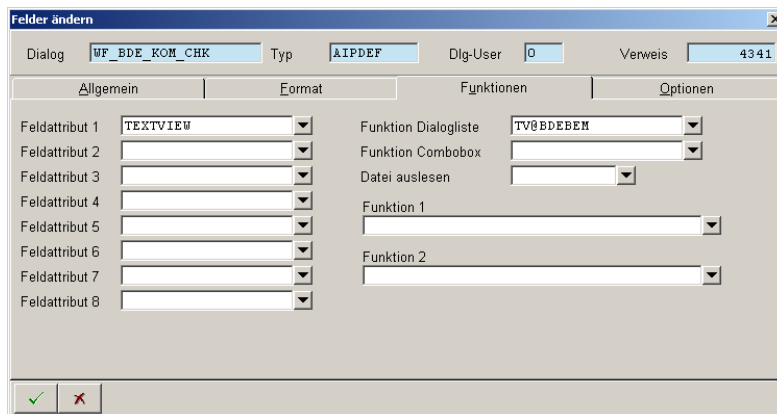
Position:
 Text X Y
 Feld X Y
 Einh. X Y

Feld-Nr.
 Text
 Einheit
 Info
 Kennung
 Kennung Index

✓ ✗

"Format" tab: selection of "grid"

"Functions" tab:



The field attribute "TEXTVIEW" defines the exact type of the "grid" pre-selection.

The field "dialog list function" includes the section of the ctaipplay.ini file in which further configurations are defined.

Ctaipplay.ini:

```
[TV@BDEBEM]
CMD=DLG=LIST; 61 | MNR=<MNR> | ANR=<ANR> |
FILE=BdeBem.lst
TEXTFILE=BdeBem.txt
CONVERSION=BDEBEM
FONTNAME=Arial
FONTSIZE=10
```

CMD: Command for loading the list. The placeholders <ANR>, <MNR> are replaced by the current values. The values must be available in the dialog and pre-assigned to default values.

FILE: This file is loaded by the above-mentioned command

TEXTFILE: This file is displayed

CONVERSION: Rule for the conversion from FILE to TEXTFILE (up to now BDEBEM available only). The file is only copied if nothing is entered.

FONTNAME, FONTSIZE: configuration of the font in the TextViewer